# Instabilities in hydrodynamics and MHD

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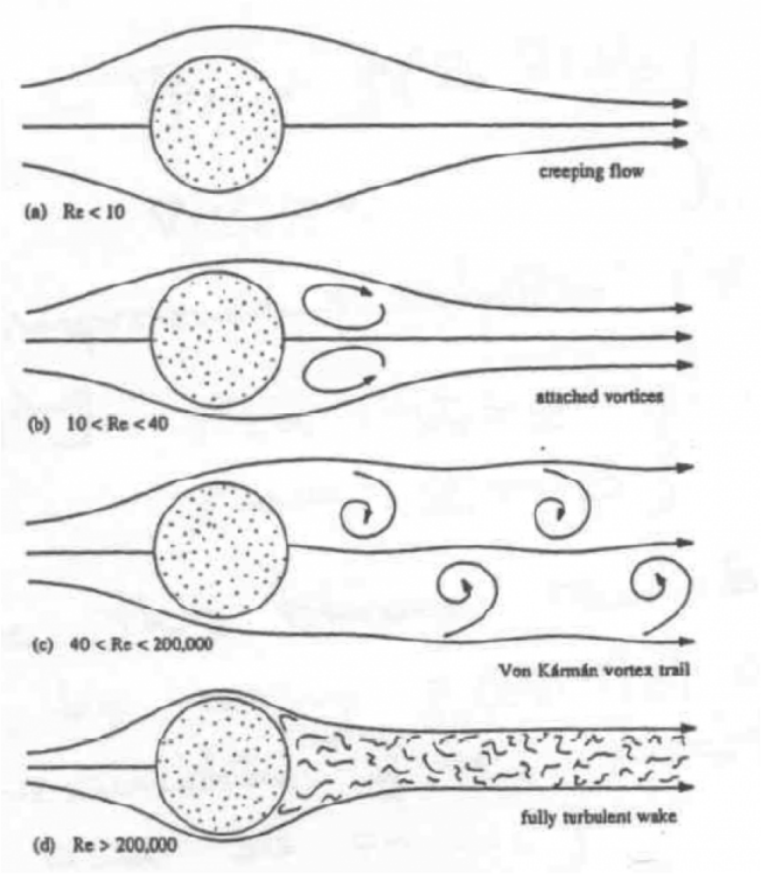
# Why do we need them?

See my previous talk on this seminar: we need a source for turbulence. If the flow is stable, there will be no turbulence.

$$\mathrm{Re} = \frac{VL}{\nu}$$

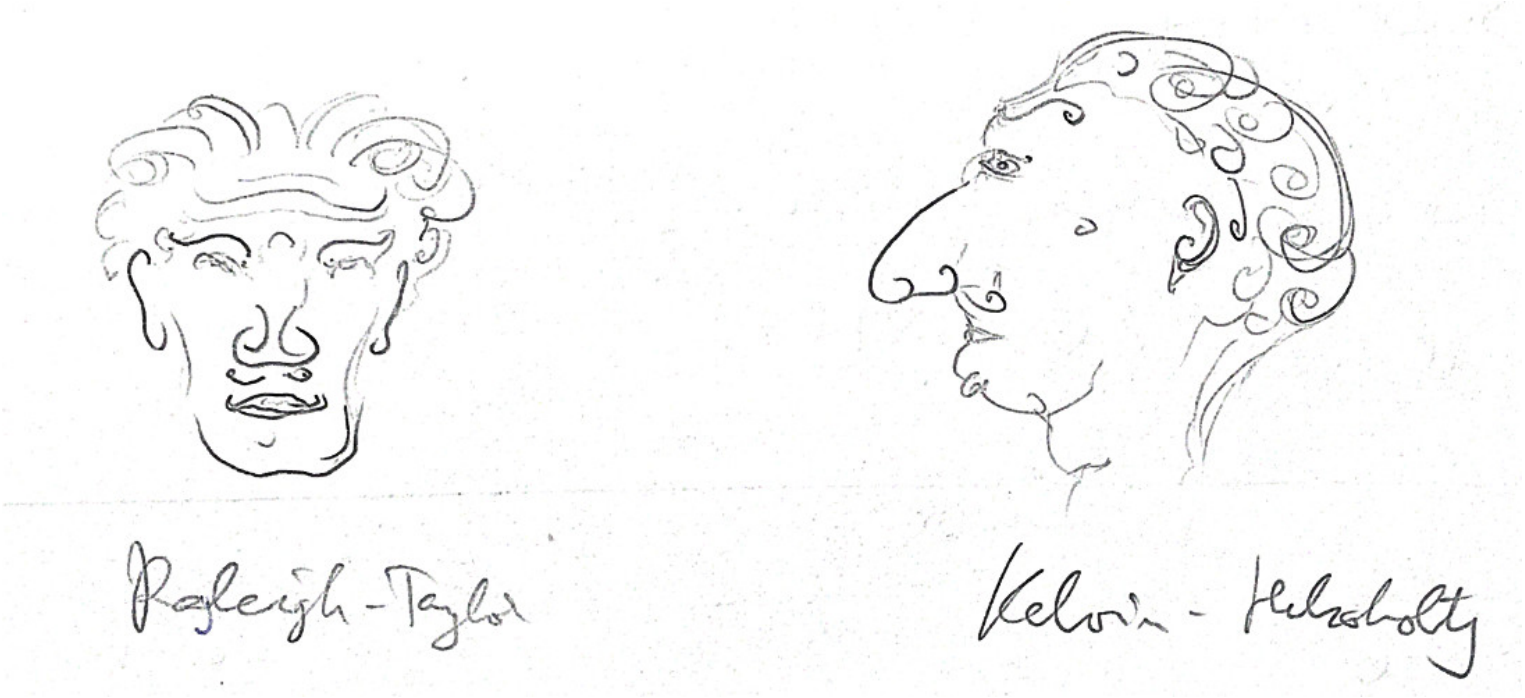
In a HD flow, chaotic velocity field is created, in an unstable MHD flow, magnetic fields are amplified as well

$$\mathrm{Re_M} = \frac{4\pi\sigma}{c^2} VL$$



# Discontinuity instabilities

Rayleigh-Taylor and Kelvin-Helmholtz



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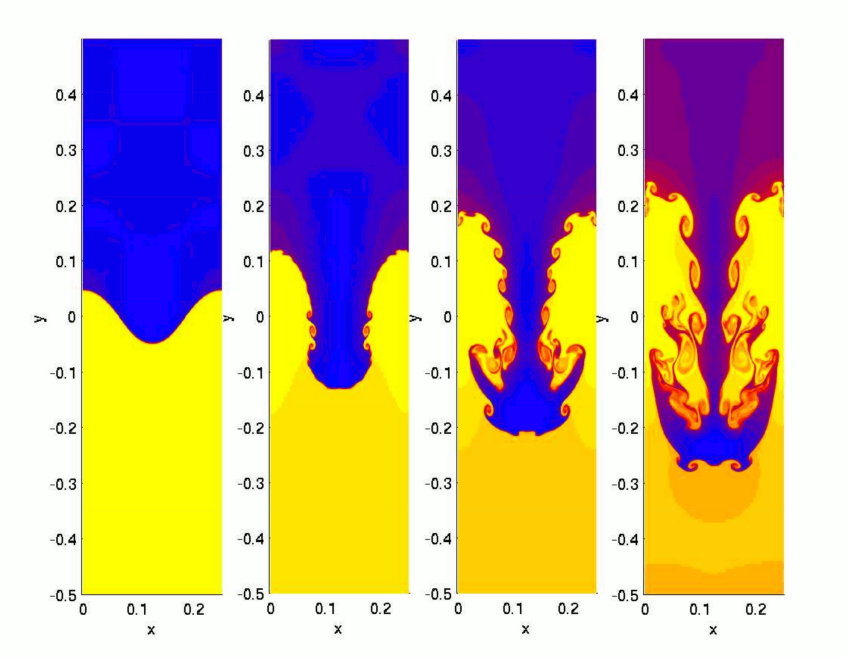


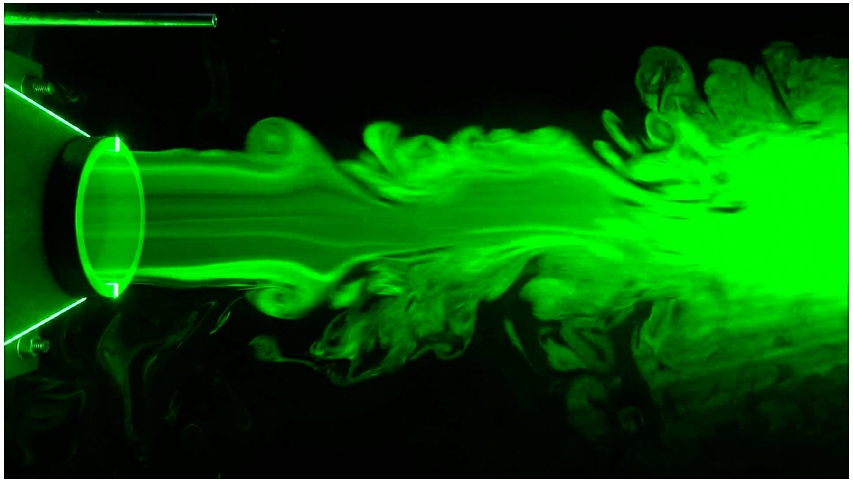

$\Delta\rho(\mathbf{n} \cdot g_{\mathrm{eff}})$

(or Richtmyer-Meshkov)

$\Delta\mathbf{v} \times \mathbf{n}$

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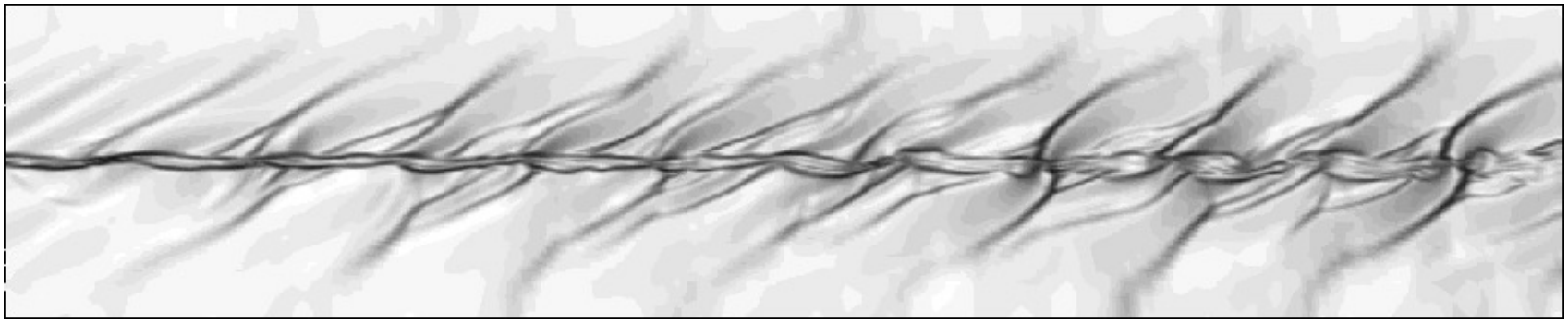

– supersonic KH at Mach around 5

# The general smooth HD problem

Small time-dependent perturbations of some stationary flow

$$\mathbf{v}(\mathbf{x}, t) = \mathbf{v}_0(\mathbf{x}) + \delta\mathbf{v}(\mathbf{x}, t)$$

$$\rho(\mathbf{x}, t) = \rho_0(\mathbf{x}) + \delta\rho(\mathbf{x}, t)$$

$$p(\mathbf{x}, t) = p_0(\mathbf{x}) + c_{\mathrm{s}}^2(\mathbf{x})\delta\rho(\mathbf{x}, t)$$

Continuity:

$$\frac{\partial}{\partial t}(\rho) + \nabla \cdot (\rho\mathbf{v}) = 0$$

Momentum (Euler):

$$\frac{\partial}{\partial t}(\mathbf{v}) + (\mathbf{v} \cdot \nabla)\mathbf{v} = -\frac{1}{\rho}\nabla p - \mathbf{g}$$

# Evaluating the problem

Zeroth order:

$$\nabla \cdot (\rho_0 \mathbf{v}_0) = 0$$

$$\underbrace{(\mathbf{v}_0 \nabla)\mathbf{v}_0}_{\text{inertial forces, } \ldots} = -\frac{\nabla p_0}{\rho_0} - \mathbf{g}$$

First order:

$$\frac{\partial}{\partial t}(\delta\rho) + \nabla \cdot (\mathbf{v}_0 \delta\rho + \rho_0 \delta\mathbf{v}) = 0$$

$$\frac{\partial}{\partial t}(\delta\mathbf{v}) + (\delta v \cdot \nabla)\mathbf{v}_0 + (\mathbf{v}_0 \cdot \nabla)\delta v = -\frac{1}{\rho_0}\nabla\left(c_s^2 \delta\rho\right) + \frac{\delta\rho}{\rho^2}\nabla p_0$$

# No underlying velocity field $v_0 = 0$

$$\begin{cases} \frac{\partial}{\partial t}\left(\delta\rho\right) + \nabla \cdot \left(\rho_0 \delta \mathbf{v}\right) = 0 \\ \frac{\partial}{\partial t}\left(\delta \mathbf{v}\right) = -\frac{1}{\rho_0}\nabla\left(c_{\mathrm{s}}^2 \delta\rho\right) + \frac{\delta\rho}{\rho^2}\nabla p_0 \end{cases}$$

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Taking into account $\nabla p_0 = -\mathbf{g}\rho$,

$$\underbrace{\frac{\partial^2}{\partial t^2}(\delta\rho) - c_s^2 \nabla^2 \delta\rho}_{\text{sonic waves}} = \left(\mathbf{g} + \nabla c_s^2\right)\nabla\delta\rho + \left(\nabla^2 c_s^2 - \frac{1}{\rho_0}(\mathbf{g}\cdot\nabla)\rho_0\right)\delta\rho$$

Sonic waves is what you get for $|\nabla^2 \delta\rho| = k^2 |\delta\rho| \gg$

# Internal gravity waves

Let us consider the longest possible waves, neglecting $\nabla\delta\rho$ where possible.

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All is left is an oscillating mode:

$$N^2 = -\frac{1}{\gamma}(\mathbf{g}\nabla)\ln\frac{p}{\rho_0^\gamma}$$

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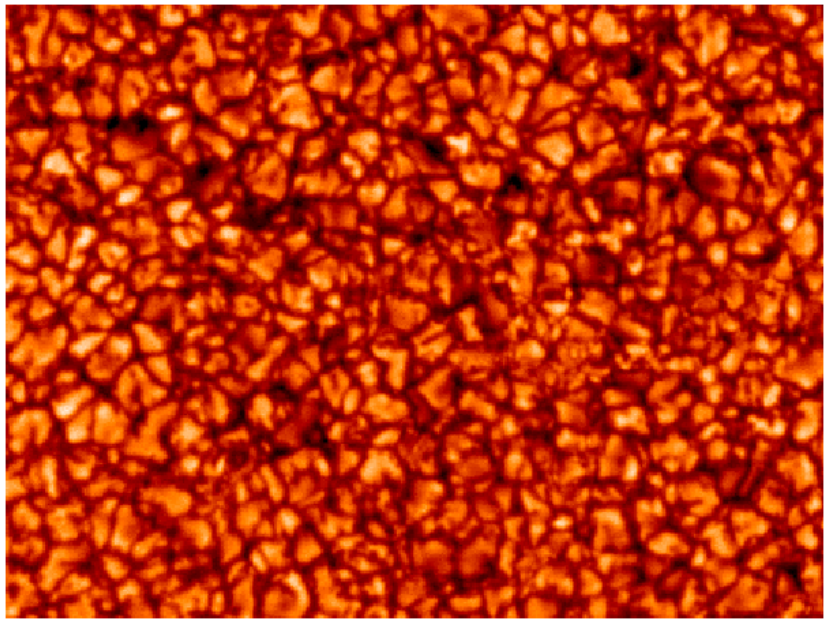
Brunt-Väisälä frequency.

# Convection

If we take into account the $\nabla^2 c_s^2$ term,

$$N^2 = -\frac{1}{\gamma}\left(\mathbf{g}\nabla\right)\ln\left(\frac{p}{\rho^\gamma}\right)$$



If entropy decreases upward. . . Schwarzschild criterion:

$$N^2 > 0$$

Does it work in accretion flows?

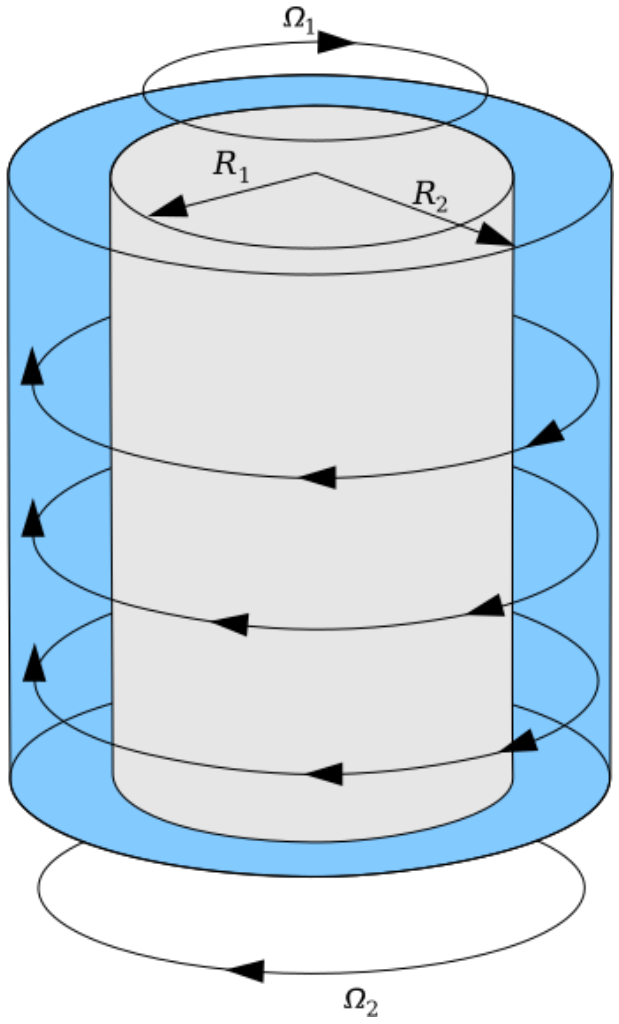
# Taylor-Couette flow

Stability of a differentially rotating axisymmetric medium. $\frac{d}{dR}\left(\Omega R^2\right) \geq 0$ – Rayleigh unstable.
Epicyclic frequency:

$$\varkappa^2 = \frac{2\Omega}{R}\frac{d}{dR}\left(\Omega R^2\right)$$

More general buoyancy-shear stability criterion:

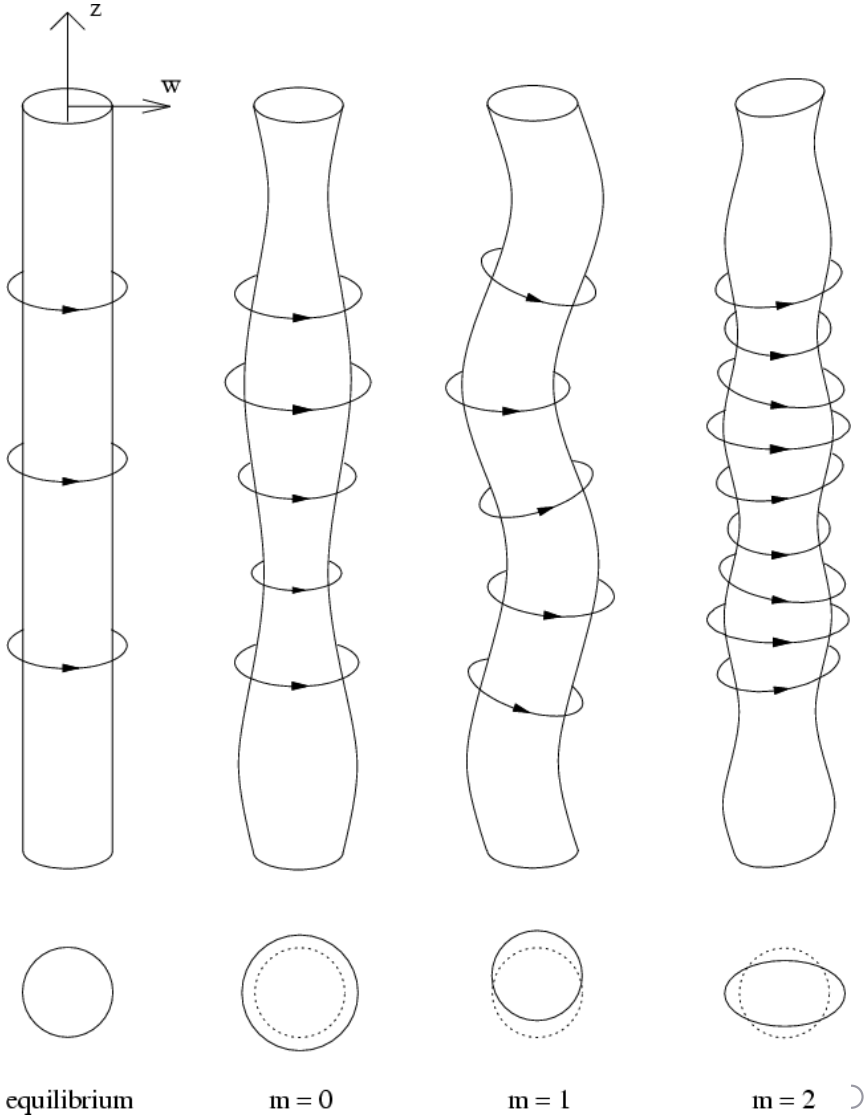
$$N^2 + \varkappa^2 \geq 0$$



# MHD instabilities

Magnetic fields are dynamically important in ionized accretion discs when

$$\frac{B_{r,z} B_{\varphi}}{4\pi} \sim p_{\text{gas, radiation}}$$



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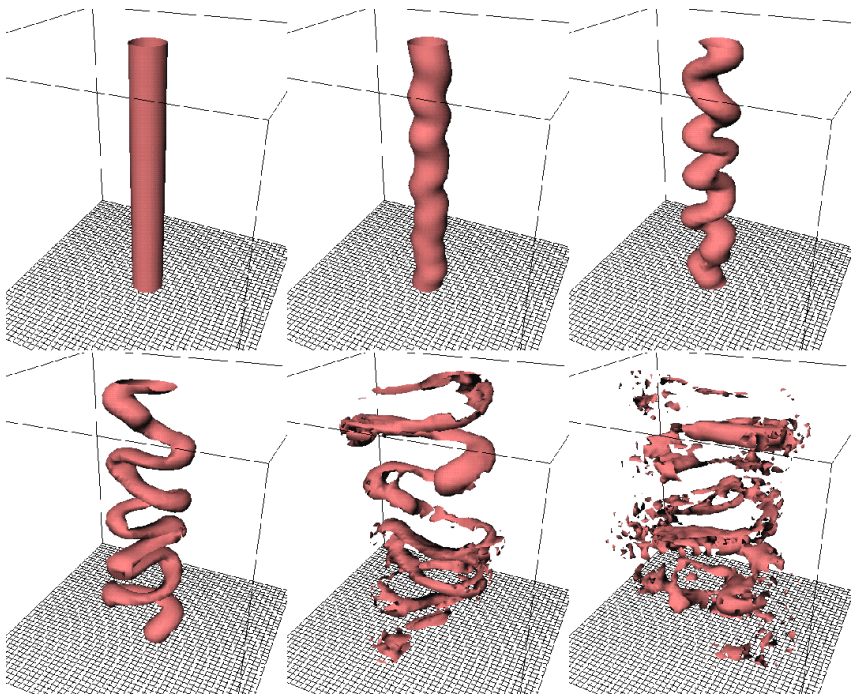


# The magnetorotational instability

Balbus & Hawley 1991

System of equations:

$$\frac{\partial}{\partial t}(\rho) + \nabla \cdot (\rho \mathbf{v}) = 0$$

$$\frac{\partial}{\partial t}(\mathbf{v}) + (\mathbf{v}\nabla)\mathbf{v} = -\frac{1}{\rho}\nabla\left(p + \frac{B^2}{8\pi}\right) + \frac{1}{4\pi\rho}(\mathbf{B}\nabla)\mathbf{B} - \mathbf{g}$$

$$\frac{\partial}{\partial t}(\mathbf{B}) = [\nabla \times [\mathbf{v} \times \mathbf{B}]]$$

WKB analysis:

$$\tilde{\omega}^4 + \frac{1}{k^2}\left(k_z^2 \varkappa^2 + (k_r N_z - k_z N_r)^2\right)\tilde{\omega}^2 - 4\Omega^2 \frac{k_z^4}{k^2} v_{\mathrm{A}}^2 = 0$$

where

$$\tilde{\omega}^2 = \omega^2 + k_z^2 v_{\mathrm{A}}^2$$

$$\varkappa^2 = \frac{2\Omega}{R}\frac{d}{dR}\left(\Omega R^2\right)$$

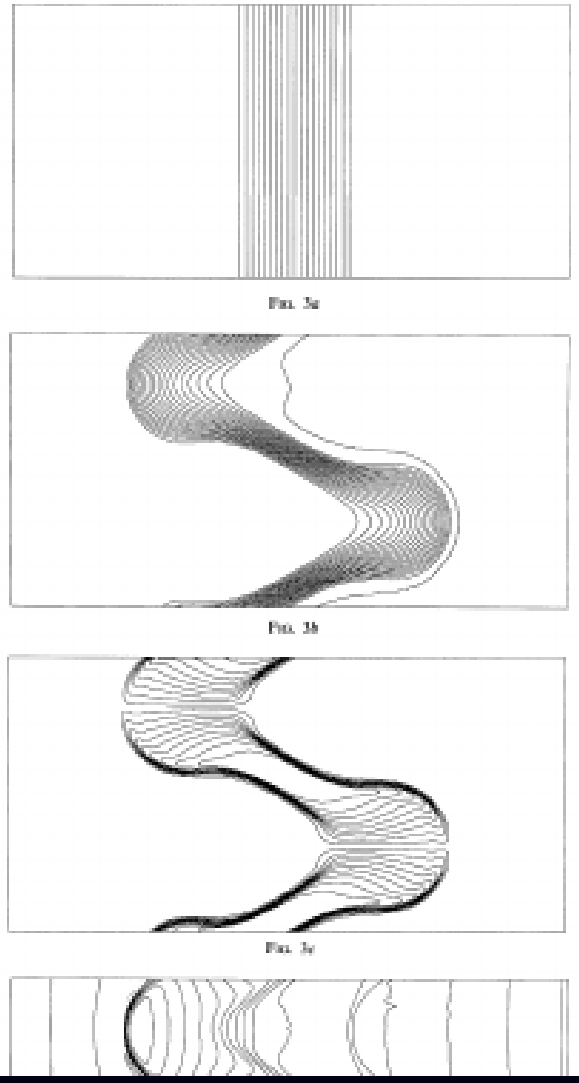
$$N_{r,z}^2 = -\frac{1}{\gamma} g_{(r,z)\mathrm{eff}} \frac{\partial}{\partial r, z}\left(\ln\left(\frac{p}{\rho^\gamma}\right)\right)$$

$$v_{\mathrm{A}} = \frac{B}{\sqrt{4\pi\rho}}$$

Solutions exist iff $\frac{d}{dr}\Omega^2 < 0$

# Nonlinear MRI

Vertical, large-scale vertical MF is converted to smaller-scale, $\sim 1/k_z$, structures dominated by $B_r$ and $B_\varphi$.
Dominated by $B_\varphi$ (amplified by differential rotation).



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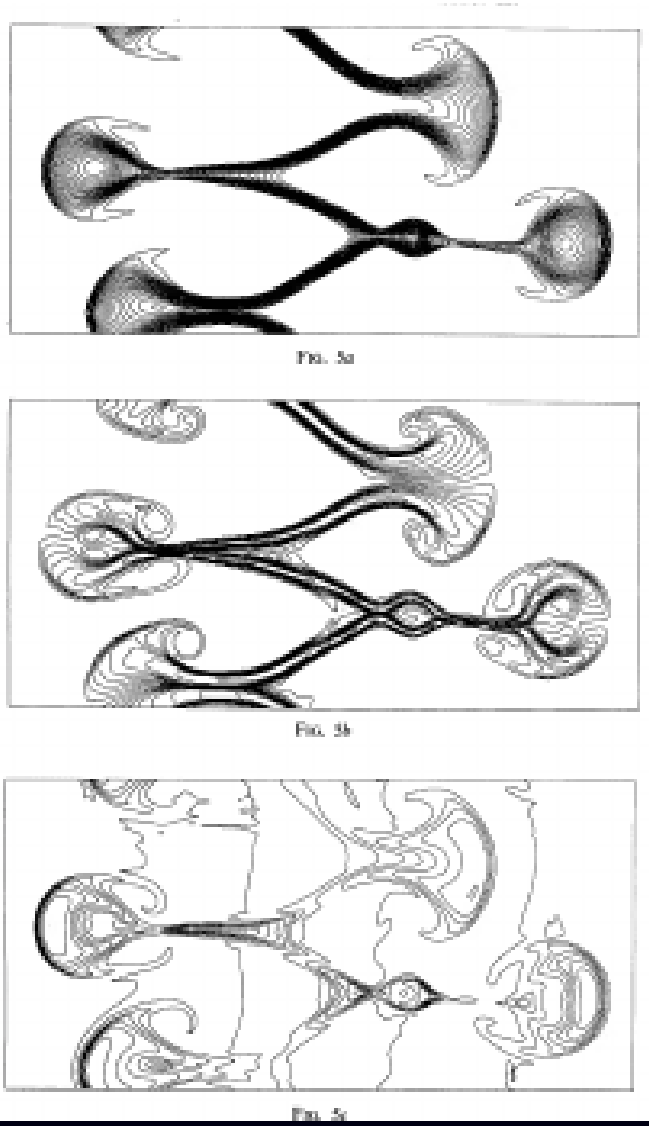
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